

2022-Ph-H1-Q2

Section: Our Dynamic Universe

Topic: Motion, Equations and Graphs

Which of the following pairs shows correct components of velocity for an object projected at an angle?

Answer: A

(Horizontal component = $u \cos \theta$, Vertical component = $u \sin \theta$)

Guidance for Students:

Breaking a vector into components is essential for projectile motion. The horizontal component uses cosine, the vertical uses sine — think of how adjacent and opposite sides relate to the angle.

Revision Tips:

- Horizontal: $v_x = u \cos \theta$
- Vertical: $v_y = u \sin \theta$
- Draw a triangle to help visualise directions.